

ASSIGNMENT 3.1

Switch and Router Configuration Basics Theory

Task (1):

Creating VLANs 10 and 20 on the switch

Switch (config) #vlan 10

Switch (config-vlan) #name Sales

Switch (config-vlan) #exit

Switch (config) #vlan 20

Switch (config-vlan) #name Support

Switch (config-vlan) #exit

Task (2):

Assign Ports to VLANs in Cisco Packet Tracer

Assuming you are using a 2960 switch, we will assign:

- FastEthernet 0/1, 0/2, 0/3 → VLAN 10 (Sales)
- FastEthernet 0/4, 0/5 → VLAN 20 (Support)

1. Open the Switch CLI

ENTER

enable

configure terminal

2. Assign 3 ports to Sales (VLAN 10)

interface range fastEthernet 0/1 - 3

switchport mode access

switchport access vlan 10

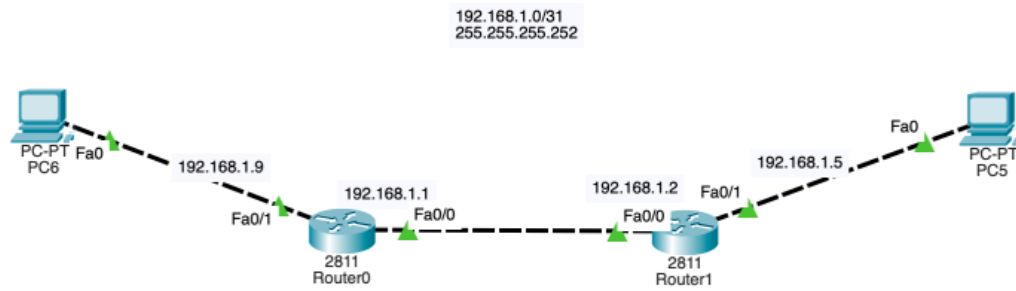
exit

3. **Assign 2 ports to Support (VLAN 20)**
 interface range fastEthernet 0/4 - 5
 switchport mode access
 switchport access vlan 20
 exit

4. **Verify VLAN Membership**
 end
 show vlan brief

VLAN	Name	Ports
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

Task (3):



Task (4):

Sales VLAN pinging Support VLAN — why it fails:

- Why the ping fails: VLANs create separate broadcast domains at Layer 2- even though both devices are physically connected to the same switch, the switch treats VLAN 10 and VLAN 20 as two entirely separate logical networks. A switch alone only forwards traffic within the same VLAN; it has no Layer 3 routing capability to move traffic between different VLANs, so the ping simply never leaves the Sales VLAN's broadcast domain.